AITS
ALL INDIA TEST SERIES

Dropper JEE Advanced

AJPM9.1/08

Test- 08

DURATION : 180 Minutes | **DATE : 16/02/2025** | **M.MARKS : 180**

Topics covered

Physics:	Full Syllabus
Chemistry:	Full Syllabus
Mathematics:	Full Syllabus

General Instructions:

1. Immediately fill in the particulars on this page of the test booklet.
2. The test is of **3 hours** duration.
3. The test booklet consists of 54 questions. The maximum marks are **180**.

SECTION-1 (Maximum Marks: 24)

This section contains **EIGHT (08)** questions.

- The answer to each question is **NUMERICAL VALUE**.
- For each question, enter the correct integer corresponding to the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer. If the numerical value has more than two decimal places, **truncate/round-off** the value of **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +3 If **ONLY** the correct integer is entered;

  Zero Marks : 0 In all other cases.

SECTION-2 (Maximum marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme:

  Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;

  Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;

  Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;

  Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;

  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

  Negative Marks : –2 In all other cases.

SECTION-3 (Maximum marks: 12)

- This section contains **FOUR (04)** Matching List Sets.
- Each set has **ONE** Multiple Choice Question.
- Each set has **TWO** lists: **List-I** and **List-II**.
- **List-I** has Four entries (I), (II), (III) and (IV) and **List-II** has **Five entries** (P), (Q), (R), (S) and (T).
- **FOUR** options are given in each Multiple Choice Question based on **List-I** and **List-II** and **ONLY ONE** of these four options satisfies the condition asked in the Multiple Choice Question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 **ONLY** if the option corresponding to the correct combination is chosen;

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : –1 In all other cases.

OMR Instructions:

1. Use blue/black dark ballpoint pens.
2. Darken the bubbles completely. Don't put a tick mark or a cross mark where it is specified that you fill the bubbles completely. Half-filled or over-filled bubbles will not be read by the software.
3. Never use pencils to mark your answers.
4. Never use whiteners to rectify filling errors as they may disrupt the scanning and evaluation process.
5. Writing on the OMR Sheet is permitted on the specified area only and even small marks other than the specified area may create problems during the evaluation.
6. Multiple markings will be treated as invalid responses.
7. **Do not fold or make any stray mark on the Answer Sheet (OMR).**

Name of the Students (In CAPITALS) : _____

Roll Number : _____

OMR Bar Code Number : _____

Candidate's Signature : _____ Invigilator's Signature _____

IMPORTANT CONSTANTS

Speed of light in free space,	:	$3.00 \times 10^{8}\ ms^{-1}$
Permeability of free space,	:	$4\pi \times 10^{-7}\ Hm^{-1}$
Permittivity of free space,	:	$8.85 \times 10^{-12}\ Fm^{-1}$
The Planck constant,	:	$6.63 \times 10^{-34}\ Js$
Rest mass of electron,	:	$9.1 \times 10^{-31}\ kg$
Rest mass of proton,	:	$1.67 \times 10^{-27}\ kg$
Molar gas constant,	:	$8.31\ JK^{-1}\ mol^{-1}$
The Avogadro constant,	:	$6.02 \times 10^{23}\ mol^{-1}$
The Boltzmann constant,	:	$1.38 \times 10^{-23}\ JK^{-1}$
Gravitational constant,	:	$6.67 \times 10^{-11}\ N\ m^2kg^{-2}$
Acceleration of free fall	:	$9.8\ ms^{-2}$
Rydberg Constant	:	$1.097 \times 10^{7}\ m^{-1}$
Atomic mass unit	:	$1.67 \times 10^{-27}\ kg$
Charge on proton	:	$1.6 \times 10^{-19}\ C$

IMPORTANT VALUES

$\sqrt{2}$ = 1.414

$\sqrt{3}$ = 1.732

$\sqrt{5}$ = 2.236

π = 3.142

e (Euler's constant) = 2.718

ln 10 = 2.303

$\log_{10}2$ = 0.3010

$\log_{10}3$= 0.4770

$\log_{10}7$ = 0.845

**** Use above values unless otherwise specified in a question*.**

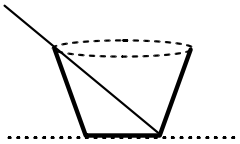
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PART-I (PHYSICS)

SECTION-1

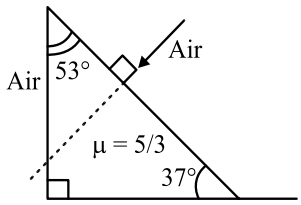
Numerical Value Type Questions:

1. A cup having shape of a frustum of a cone is placed on a horizontal tabletop. Mass of the cup is m = 40 g and diameter of its bottom is d =10 cm. A uniform rod of mass m = 20 g is placed in the cup as shown. The rod is inclined at an angle $\theta = 60°$ to the horizontal. Find maximum length l (in m) of the rod so that the system will not turn over.

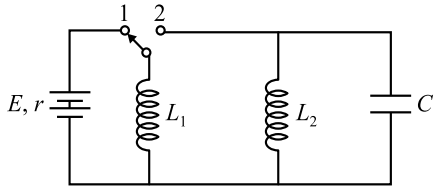


2. Thermal conductivity in wood depends on the direction of heat flow. In a wood sample, thermal conductivity along the fibres is $\eta = 2$ times of that perpendicular to the fibres. Two identical cylinders are cut from this wood; axis of the first cylinder is parallel to the fibres and that of the second makes an angle $\theta = 37°$ with the fibres. The curved surfaces of both the cylinders are well insulated from the surroundings. If equal temperature differences are applied across their circular ends, find ratio of heat current in the second cylinder to that in the first cylinder.

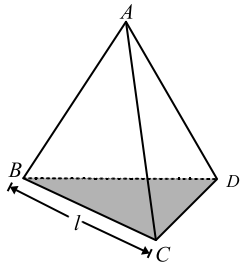
3. A ray of light falls perpendicularly on the face of a right triangular prism as shown in the figure. If the total deviation suffered by the ray in degrees is δ. Find $\frac{\delta}{100}$. (Refractive index of prism material is $\frac{5}{3}$)



4. A circuit shown consists of two inductors of inductances L_1 and L_2, a capacitor of capacitance C, a battery of electromotive force E and internal resistance r and a switch. Initially the switch was in position 1 for a long time. Find the maximum current (in A) in the inductor L_2 after the switch is thrown to position 2?
(Given: $E = 5V$, $L_1 = 4$ mH, $L_2 = 1$ mH, $r = 2\Omega$)

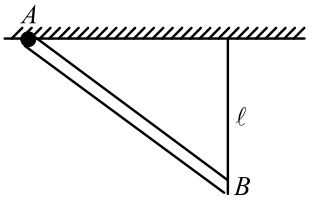


5. An insulating hollow Regular Tetrahedron of each equilateral triangular face of side l carries uniform charge on all of its surfaces. The surface charge density is σ. The force acting on a face due to all the other faces is $\frac{\sigma^2 l^2}{x\varepsilon_0}$. Find the value of $\sqrt{3}x$?

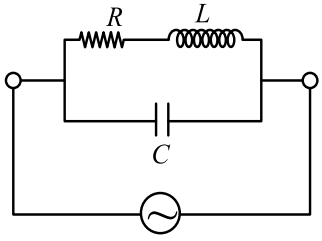


6. A string of mass per unit length 0.8 kg/m is stretched to a tension 500 N. Find the mean power (in watt) required to maintain a travelling wave of amplitude of 10 mm and wavelength 0.5 m. (Take $\pi^2 = 10$)

7. A thin uniform rod is hinged at one end A and the other end B is supported by a vertical massless inextensible thread of length l as shown in the figure. Now the end B is slightly displaced out of the plane such that thread always remains taut. The time period of small angular oscillations of this system is $2\pi\sqrt{\frac{2l}{xg}}$, where g is the acceleration due to gravity. Find the value of x.



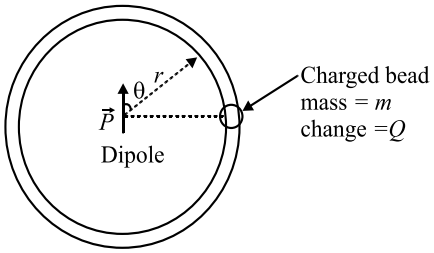
8. Consider the circuit shown in the figure. The resonance frequency for the circuit is $50x$ Hz. Find the value of x. (Use $R = 2\Omega$, $L = 1$ mH, $C = 8$ μF and $\pi = 22/7$)



SECTION-2

One or More Than One Correct Type Questions:

9. A small charged bead can slide on a circular frictionless, insulating wire frame. A point like dipole of dipole moment $\vec{P}$ is fixed at the centre of circle. Initially, the bead is on the plane of symmetry of the dipole. Bead is released from rest. Ignore the effect of gravity. Select the correct options: (θ-angle made by position vector with dipole moment)



(A) Magnitude of velocity of bead as a function of its angular position $\sqrt{\frac{-QP\cos\theta}{2\pi\varepsilon_0 mr^2}}$

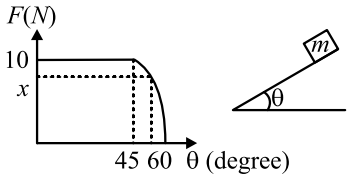
where $\left\{\frac{\pi}{2} \le \theta \le \pi\right\}$

(B) Normal force exerted by the wire frame on bead is zero.

(C) If the wire frame were not present, bead executes to and fro oscillatory circular motion and keeps returning back to initial point

(D) Bead executes periodic motion.

10. An object of mass *m* is kept on a rough inclined surface ($\mu_s = \mu_k$). If angle of inclination is gradually increased, the net contact force F_C, offered by surface on the object vary with incline angle θ, as shown in the graph (g = 10 m/s^2). Choose the correct option(s):



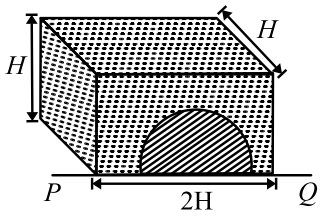
(A) Mass of object is 1 kg

(B) Coefficient of friction is $\sqrt{3}$

(C) Coefficient of friction is 1

(D) Value of *x* is $5\sqrt{5}$ N

11. A liquid of density 'ρ ' is filled in a cuboidal container of dimension ($H \times H \times 2H$) as shown in the figure. A semicircular gate is made at the bottom of container of radius $R = \frac{H}{2}$. Let *F* is the force acting on the gate due to the liquid and τ is the torque acting on the gate due to the liquid about a line *PQ*. Then choose the correct option(s).



(A) $F = \frac{H^3\rho g}{4}\left(\frac{\pi}{2} - \frac{1}{3}\right)$

(B) $F = \frac{H^3\rho g}{2}\left(\frac{\pi}{4} - \frac{1}{6}\right)$

(C) $\tau = \frac{H^4\rho g}{4}\left(\frac{1}{3} - \frac{\pi}{32}\right)$

(D) $\tau = \frac{H^4\rho g}{3}\left(\frac{1}{3} - \frac{\pi}{16}\right)$

12. When a particle of mass *m* is restricted to move along *x* -axis between *x* = 0 and *x* = *a*, where *a* is of nanometre dimension, its energy can take only certain specific values. The allowed energies of the particle moving in such a restricted region, correspond to the formation of standing waves with nodes at its ends *x* = 0 and *x* = *a*. The wavelength of this standing wave λ is related to the linear momentum as $p = \frac{h}{\lambda}$ of the particle according to the de-Broglie relation. The energy of the particle can be denoted by a quantum number '*n*' taking value 1, 2, 3,____ (*n* = 1, called the ground state) corresponding to the number of loops in the standing wave.

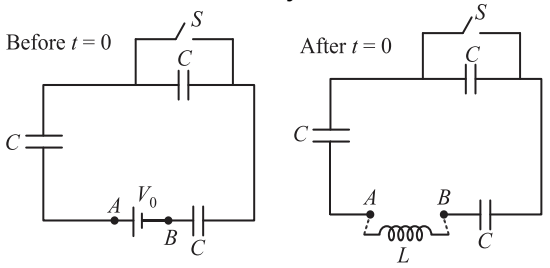
(A) The allowed energy of the particle for a particular value of *n* is proportional to a^{-1}

(B) The allowed energy of the particle for a particular value of n is proportional to a^{-2}

(C) If the mass of the particle is $m = 1.0 \times 10^{-30}$ kg and *a* = 6.6 nm, the energy of the particle in its ground state is approximately equal to 8×10^{-3} eV

(D) If the mass of the particle is $m = 1.0 \times 10^{-30}$ kg and *a* = 6.6 nm, the energy of the particle in its ground state is approximately equal to 8×10^{-5} eV

13. Three capacitors each of capacitance C are connected in series as shown. Initially the switch S is open. Now capacitors are charged by a battery of emf V_0 by connecting across A and B. At $t = 0$ battery is disconnected and inductor of inductance L is connected between A and B. After then, switch S is closed at any further time t_0.



(A) If $t_0 = 2\pi\sqrt{\frac{LC}{3}}$ then after closing the switch S, energy loss in the system is $\frac{CV_0^2}{18}$

(B) If $t_0 = 2\pi\sqrt{\frac{LC}{3}}$ then after closing the switch S, energy loss in the system is $\frac{CV_0^2}{36}$

(C) If $t_0 = \frac{\pi}{3}\sqrt{\frac{LC}{3}}$, after closing the switch S, Q_{max} be charge amplitude, then $Q_{max} = \frac{CV_0}{6}\sqrt{\frac{11}{2}}$

(D) If $t_0 = \frac{\pi}{3}\sqrt{\frac{LC}{3}}$, after closing the switch S, Q_{max} be charge amplitude, then $Q_{max} = \frac{CV_0}{2}\sqrt{\frac{11}{6}}$

14. There are two concentric metallic spherical shells of radii a and $b(a < b)$. The shells are being maintained at a constant potential difference V. The medium between the shells is filled with a material of resistivity ρ. At a distance $r(a < r < b)$ from the centre of the shells

(A) Electric field is $\frac{Vab}{r^2(b-a)}$

(B) Current density is $\frac{Vab}{\rho r^2(b-a)}$

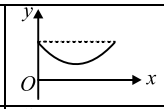
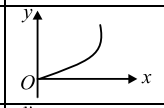
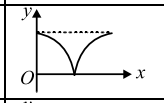
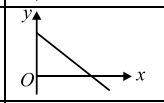
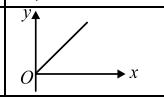
(C) Electric field is $\frac{V}{(b-a)}$

(D) Current density is $\frac{V}{\rho(b-a)}$

SECTION-3

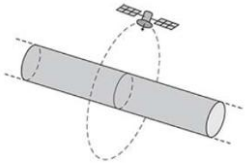
Matrix Match Type Questions:

15. A small body is projected up at an angle $\theta(<90°)$ to the horizontal. List-I contains variables to be taken on x and y - axes and list-II contains corresponding graphs for this body, match the following. [The graph shapes are not to the scale, choose the best possible qualitative match]

List -I		List -II	
I	Slope of trajectory of the body on y-axis and time on x- axis	P	
II	Magnitude of tangential acceleration on y- axis and time on x-axis	Q	
III	Magnitude of normal acceleration on y- axis altitude h on x-axis	R	
IV	Speed of projectile on y- axis and time on x- axis	S	
		T	

	I	II	III	IV
(A)	S	R	P	Q
(B)	S	R	Q	P
(C)	T	Q	P	S
(D)	P	Q	T	S

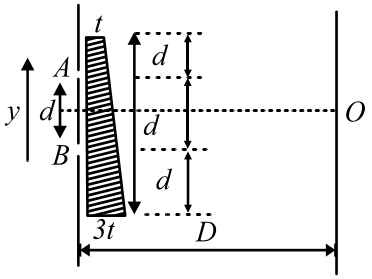
16. One of the planets of a star called 'Noname' has the shape of a long cylinder. The average density of the planet is the same as that of the Earth, its radius R is equal to the radius of the Earth, and the period of its rotation around its long axis is just one day. (Take $R = 6400$ km, g = 10 m/s^2 and ln(2) = 0.7) (Assume that Geostationary Satellite is defined as a satellite that appears to be at a fixed point in space as it appears from a planet's surface)



List-I		List-II	
I	The speed (in km/sec) of a satellite in a stable orbit just above the planet's surface is	P	1.4
II	The ratio of the altitude of a 'geostationary' communications satellite above the surface to the radius of the planet	Q	2.8
III	The orbital velocity (in km/sec) of 'geostationary' satellite is	R	20.05
IV	The ratio of the energy required to double its orbital radius to its kinetic energy is	S	9.8
		T	19.6

	I	II	III	IV
(A)	P	S	T	Q
(B)	S	P	S	T
(C)	S	R	S	P
(D)	Q	R	P	S

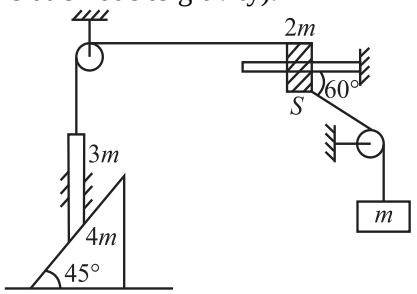
17. In a standard *YDSE* two slits are symmetrically covered by a wedge shape film of refractive index μ as shown in the figure. The thickness of film varies linearly from *t* to 3*t* from one end as shown in the figure. The distance between the two slits is *d*. The screen is at distance *D* from slits. Screen is sufficiently large enough but fringes at infinity can not be observed. (Figure not drawn upto the scale)
(Given d = 1 mm, t = 2 mm, λ = 60 μm, μ=1.5, D = 3 m)



List -I		List -II	
I	Central maxima is formed at a distance x (in m) from O then the value of $\sqrt{180}x$ is	P	32
II	Total number of bright fringes observed on the screen is	Q	33
III	If the film is displaced by d along -y axis such that slit A is just uncovered then number of bright fringes observed on the screen is	R	34
IV	If the film is displaced by 1.5d along +y axis such that slit B is uncovered than number of bright fringes observed on the screen is	S	35
		T	36

	I	II	III	IV
(A)	T	Q	P	S
(B)	P	Q	S	T
(C)	T	Q	Q	R
(D)	P	S	R	Q

18. Consider an arrangement shown in the figure. The rod of mass 3*m* is constrained to move in the vertical direction only. The wedge of 4*m* moves in the horizontal direction. The slider *S* of mass 2*m* moves on a fixed horizontal rod. At a particular instant system is released and the string connecting the slider *S* to the block of mass *m* making an angle 60° with the rod. (Neglect friction everywhere) Match the statements from List I with those in List II and select the correct answer using the code given below the lists. (g is acceleration due to gravity).



List- I		List-II	
I	If net force on the block of m at this instant is $\left(\frac{5mg}{37}\right)a$, then the value of a is	P	3
II	If net force on the wedge is $\left(\frac{5mg}{37}\right)b$, then the value of b is	Q	1
III	If contact force between wedge and rod is $\left(\frac{5\sqrt{2}mg}{37}\right)c$, then the value of c is	R	2
IV	If tension in the string between blocks having mass m and 2m is $\left(\frac{21mg}{37}\right)d$, then the value of d is	S	8
		T	4

	I	II	III	IV
(A)	P	Q	T	S
(B)	Q	S	S	R
(C)	Q	P	R	Q
(D)	R	S	P	T

PART-II (CHEMISTRY)

SECTION-1

Numerical Value Type Questions:

19. In 200 ml solution of 0.1 M NaOH and 0.2 M NH_4OH, 300 ml of 0.1 M HCl is added. Find pH of solution (pK_b of NH_4OH = 5, log3 = 0.48).

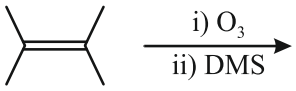
20. When one mole of $K_4[Fe(CN)_6]$ is heated with conc. H_2SO_4, the evolved gas during reaction is reacted with excess of I_2O_5. How many moles of I_2 will be liberated during the reaction?

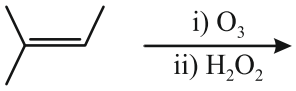
21. Consider the following complexes:

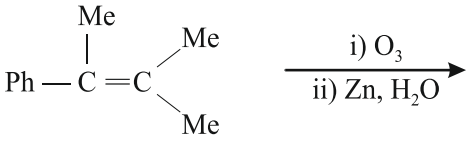
Compound	No. of stereoisomers
$[Co(NH_3)_3Cl_3]$	x
$[Pt(Cl)(Br)(H_2O)_2]$	y
$[Co(C_2O_4)_3]^{3-}$	z

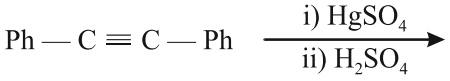
The value of (x + y + z) is _____.

22. Total number of reactions in which product gives positive Tollen's test, is_____.

(i) 

(ii) 

(iii) 

(iv) 

(v) 

(vi) Sucrose $\xrightarrow{H^+/H_2O}$

23. Vapour pressure of a solution containing 12 g of a non-volatile solute in 360 g water is 20 torr. If 2 moles of water is further added, then vapour pressure increases by 0.02 torr. The molar mass (in g/mol) of the non-volatile solute is_____. (consider solute as a non-electrolyte) [Round off upto nearest integer]

24. A flask is filled with N_2O_3 gas at 2 bar pressure, and following equilibrium is established.

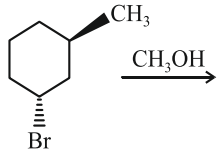
Reaction 1: $N_2O_3(g) \rightleftharpoons NO_2(g) + NO(g)$;

$K_{P_1} = 2.5$

Reaction 2: $2NO_2(g) \rightleftharpoons N_2O_4(g)$; $K_{P_2} = ?$

If at equilibrium partial pressure of NO (g) is 1.5 bar, then find equilibrium constant for second reaction, (round off upto two decimal places).

25. If X represents total number of substitution and elimination product(s), then find the value of X.



26. The specific conductivity of a saturated solution of AgCl is 2.30×10^{-6} mho cm^{-1} at 25°C. If $\lambda^{\circ}_{Ag^+}$ = 61.9 mho cm^2 mol^{-1} and $\lambda^{\circ}_{Cl^-}$ = 76.3 mho cm^2 mol^{-1} then solubility of silver chloride (in order of 10^{-3} g/L^{-1}) at 25°C, is_____. [Round off upto two decimal places] (Given, molar mass of Ag-108, (Cl – 35.5 g/mol))

SECTION-2

One or More Than One Correct Type Questions:

27. The freezing point of solution containing 0.522 g of compound 'x' (M. wt. 152 g/mol) and 0.0386 g of an unknown non-electrolyte (y) compound is 430 K. Given, the melting point of 'x' = 450 K.

ΔH_{fus} 'x' = 8.53 kJ/mol

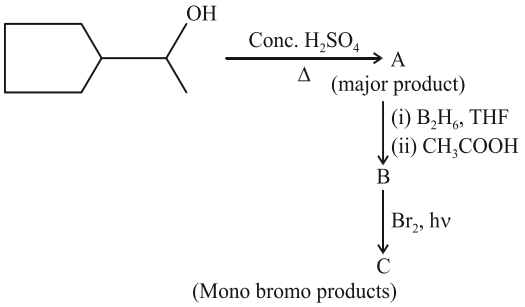
(A) K_f of 'x' = 30 K kg/mol

(B) Molality of solution is 0.667 mol/kg

(C) The molecular weight of solute = 0.157 kg/mol (nearly)

(D) The molecular weight of solute = 0.111 kg/mol (nearly)

28. Consider the given reaction sequence :



The incorrect statement(s) regarding the above reaction sequence is/are

(A) Number of fractions obtained from fractional distillation of C is 8.

(B) A on reaction with Br_2/CCl_4 can give three stereoisomers.

(C) Total number of mono-bromo products is 14.

(D) B is optically active.

29. Among the given anions with which anion $AgNO_3$ gives white precipitate?

(A) $S_2O_3^{2-}$ (B) CH_3COO^-

(C) $C_2O_4^{2-}$ (D) Cl^-

30. Which of the following statement(s) is/are correct?

(A) Angular function for s-orbital is independent of θ & ϕ.

(B) ψ^2 represents probability density of electron at a point.

(C) Total number of radial nodes for 4p orbital is 2.

(D) Radial probability density at the nucleus is zero for all the orbitals.

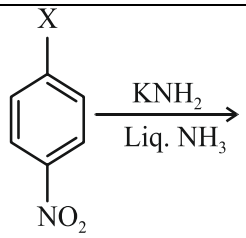
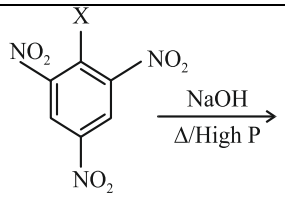
31. In the following gas phase first order reaction:
$A(g) \rightarrow 2B(g) + C(g)$
Initial pressure was found to be 400 mm of Hg and it changed to 1000 mm Hg after 20 min.
The correct statement(s) is/are (Given $10^{0.9} = 8$)
(A) Half life of A is 10 min.
(B) Rate constant is 0.0693 min^{-1}
(C) Partial pressure of C at 30 min is 350 mm Hg
(D) Total pressure after 30 min is 1150 mm Hg

32. When 5 mL of a 0.1 M HCl solution is mixed with 5 mL of a 0.1 M NaOH solution, temperature of solution increases by t °C. Which of the following statement(s) can be predicted from this observation?
(A) If 5 mL of 0.1 M HCl is mixed with 5 mL 0.1 M NH_3 solution, the temperature rise will be less than t°C
(B) If 5 mL 0.1 M CH_3COOH is mixed with 5 mL 0.1 M NaOH, the temperature rise will be less than t°C
(C) If 10 mL of 0.1 M HCl is mixed with 10 mL of 0.1 M NaOH, the temperature rise will be 2t°C.
(D) If 10 mL of 0.1 M HCl is mixed with 10 mL of 0.1 M NaCl, the temperature rise will be t°C.

SECTION-3

Matrix Match Type Questions:

33. Match the reactions given in List-I with their characteristics given in List-II

List -I		List -II	
I		P	Reaction follows reactivity order as Ar-F > Ar-Cl > Ar-Br > Ar-I
II		Q	Reaction mechanism may involve elimination
III	$CH_3CH_2CH(I)CH_3$ $\xrightarrow[\Delta]{\text{Alc. KOH}}$	R	Major product formation is done by nucleophilic substitution
IV	$CH_3CH_2I \xrightarrow{CH_3O^-}$	S	Major product is formed by 2nd order kinetics
		T	Reaction involves formation of transition state

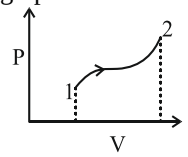
	I	II	III	IV
(A)	P,Q,R,S,T	P,R,S,T	Q,S,T	Q,R,S,T
(B)	Q,S,T	P,Q,R,S,T	P,Q,S,T	P,Q,R,S,T
(C)	Q,S,T	Q,R,S,T	P,Q,R,S,T	P,Q,R,S,T
(D)	P,Q,R	Q,R,S,T	P,Q,R,S,T	P,Q,R,S,T

34. Match the compounds given in List-I with their characteristic(s) given in List-II

List -I		List -II	
I	Chloral hydrate	P	Forms a zig-zag chain in both solid and also in the liquid state
II	Hydrogen fluoride	Q	Forms two-dimensional sheet with almost hexagonal symmetry
III	Crystalline boric acid	R	Intramolecular H-bond
IV	Sulphuric acid	S	Intermolecular H-bond
		T	No hydrogen bonding

	I	II	III	IV
(A)	R	P, S	Q, S	S
(B)	P, T	R, S	Q, S	R
(C)	R	Q, T	P, S	S
(D)	R, S	Q, R	S	P, T

35. Match list-I with list-II

List -I		List -II	
I	An insulated rigid container has two chambers, chamber I contains 1 litre of ideal gas(A) and the chamber II contains 1 litre of ideal gas (B), both at 298 K. The valve between the chambers is opened. (Assume that gases A & B are non-reacting)	P	The temperature of the gas decreases
II	An ideal monoatomic gas taken in a closed container expands to thrice its original volume such that volume $V \propto \frac{1}{\sqrt{P}}$ where P is the pressure of the gas	Q	The temperature of the gas increases or remains constant
III	An ideal monoatomic gas taken in a closed container expands to twice its original volume such that $P^2 = \frac{k}{V^3}$, where V and P are volume and pressure respectively. (k is constant)	R	The gas loses heat
IV	An ideal monoatomic gas taken in a closed container expands from 1 to 2 such that its pressure P and volume V follows the behaviour shown in the graph. 	S	The gas gains heat
		T	Entropy of universe remains constant during the process

	I	II	III	IV
(A)	Q,S,T	P,T	P	P,R,S
(B)	Q	P,R,T	P,S,T	Q,S,T
(C)	Q	S,T	P,S,T	R,S,T
(D)	T	P,R,S	P,R,T	Q,S

36. Match the List-I with List-II

List -I (Coordination Compounds)		List -II (Type of isomerism)	
I	$[Co(NH_3)_4Cl_2]Cl$	P	Optical isomerism
II	$[Co(en)_3]Cl_3$	Q	Ionization isomerism
III	$[Co(en)_2(NO_2)Cl]SCN$	R	Coordination isomerism
IV	$[Cu(NH_3)_4][PtCl_4]$	S	Geometrical isomerism
		T	Linkage isomerism

	I	II	III	IV
(A)	P,Q,T	R,T	Q,R,T	S
(B)	R	Q,S,T	S	P,Q
(C)	S,T	R,S	P,Q,R	Q,R
(D)	S	P	P,Q,S,T	R

PART-III (MATHEMATICS)

SECTION-1

Numerical Value Type Questions:

37. If $f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + ... a_nx^n + ...$ and $\frac{f(x)}{1-x} = b_0 + b_1x + b_2x^2 + ... b_nx^n + ...$

If $a_0 = 1$, $b_1 = 3$ and $a_1 = k$, then area bounded by the $y(1 + x^2) = 5 + k$, x-axis, y-axis and $x = \sqrt{3}$ is A, then $[A]$ is (where [.] denotes greatest integer function)

38. The value of $\sum_{p=1}^{10}\sum_{q=1}^{10}\sum_{r=1}^{10} p\tan^{-1}\left(\frac{q}{r}\right) = S\pi$, then $\frac{S}{100}$ is equal to

39. If $|z - (5 + 12i)| = 1$ then the value of $\frac{|z|_{min} + |z|_{max}}{5}$ is equal to

40. If $p(x) = \begin{vmatrix} 1 & 1 & 1 \\ (x+2) & (x+3) & (x+4) \\ (x+2)^2 & (x+3)^2 & (x+4)^2 \end{vmatrix}$, then value of $\frac{p(3) - p(4)}{p(5)}$ is equal to

41. Letters of the word "MOTIVATION" are arranged randomly. If the probability of these such words contains the word "IIT" always is $\frac{p}{q}$, then $p + q$ is equal to (where p and q are coprime)

42. Consider the following frequency table

Class:	10–20	20–30	30–40	40–50	50–60
Frequency	α	110	54	30	β

If the sum of all frequencies is 584 and median is 45, then $|\alpha - \beta|$ is equal to_____

43. A pair of tangents is drawn from the point A (23, 3) to the circle whose equation is $x^2 + y^2 - 2x - 4y + 4 = 0$ to touch the circle in B and C. A third tangent intersects the segment AB in P and AC in R and touch the circle in Q. If p is the perimeter of the triangle APR, than $|p - 50|$ = ___

44. Value of $\sum_{k=1}^{\infty}\sum_{r=1}^{k}\frac{1}{3^k}\left({}^kC_r\right)$ is

SECTION-2

One or More Than One Correct Type Questions:

45. Which of the following statements are INCORRECT? (where {.} denotes the fractional part of function)

(A) $f(x) = \{x\}\cos^2\left(\left(\frac{2x-1}{2}\right)\pi\right)$ is differentiable for all $x \in R$

(B) $f(x) = |\sin x|\cos^{-1}(\cos x)$ is differentiable $\forall$ $x \in (0, 2\pi)$

(C) $f(x) = ||x - 2| - |x - 6|| - 3|x| + 2x + 1$ is not differentiable at 3 points

(D) $f(x) = \{x\}|\sin \pi x|$ is not differentiable at all integers

46. If hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ has eccentricity 2 and foci coincides with the foci of ellipse $4x^2 + 9y^2 = 36$, then

(A) Length of latus rectum $5\sqrt{5}$

(B) $a^2 - b^2 = -\frac{5}{2}$

(C) $a^2 + b^2 = 16$

(D) $a^2 + b^2 = 5$

47. Let P be a point on an ellipse with eccentricity = $\frac{1}{2}$ whose minor axis is $2\sqrt{3}$ and foci S and S'. Given that $PS = 2PS'$. Then

(A) $PS = \frac{4}{3}$ (B) $PS' = \frac{4}{3}$

(C) $SS' = 2$ (D) $SS' = 4$

48. Let S be the set of all non zero real number 'a' such that the quadratic equation $ax^2 + (a-1)x + (a-2) = 0$ has two distinct real roots α and β satisfying the inequality $|\alpha - \beta| < 2$. Then which of the following interval(s) is/are a subset of S?

(A) $\left(1+\frac{2}{\sqrt{3}}, \infty\right)$ (B) $\left(1-\frac{2}{\sqrt{3}}, 1\right)$

(C) $\left(1, 1+\frac{2}{\sqrt{3}}\right)$ (D) $\left(1-\frac{2}{\sqrt{3}}, -\frac{1}{7}\right)$

49. Which of the following is/are true ?

(A) $\int_0^1 \sin\left(x^2+2x+1\right)dx - \int_1^2 \sin x^2 dx = 0$

(B) $\int_0^1 \sin(x+1)^2\, dx - \int_0^2 \sin(x)^2\, dx = 0$

(C) $\int_{-5}^{-4} \sin\left(x^2-3\right)dx + \int_{-2}^{-1} \sin\left(x^2+12x+33\right)dx = 0$

(D) $\int_{-4}^{4} \cos x^2 dx - 8\int_0^1 \cos 16(2x-1)^2\, dx = 0$

50. The solution of $\left(\frac{dy}{dx}\right)\left(x^2y^3 + xy\right) = 1$ is

(A) $1/x = 2 - y^2 + C\, e^{-y^2/2}$

(B) The solution of an equation which is reducible to linear equation.

(C) $2/x = 1 - y^2 - C\, e^{-y/2}$

(D) $\frac{1-2x}{x} = -y^2 - C\, e^{-y^2/2}$

SECTION-3

Matrix Match Type Questions:

51. Consider the triangle formed by the lines $y + 3x + 2 = 0$, $3y - 2x - 5 = 0$ and $4y + x - 14 = 0$. Match the entries of following lists and choose the correct option

List -I		List -II	
I	If (0, λ) lies inside the triangle (not boundary), then integral values that are less than $\|3\lambda\|$	P	4
II	If (1, λ) lies inside (not boundary) the triangle, then integral values that are less than $\|3\lambda\|$	Q	5
III	If (λ, 2) lies inside the triangle, then number of positive integral values of $\|6\lambda\|$ can be	R	6
IV	If (λ, 7/2) lies inside the triangle, then number of integral values of $\|6\lambda\|$ that are greater than 1 are	S	7
		T	9

	I	II	III	IV
(A)	P, Q, T	P, Q, R	S	R
(B)	R, S, T	S, R	S	S
(C)	P, Q	P, Q, R, S	S	T
(D)	Q, R, T	Q, R, S	S	P

52. A and B are two events, such that $P(A) = \frac{3}{5}$ and $P(B) = \frac{2}{3}$. Match the following lists and choose the correct option

List -I		List -II	
I	Range of $P(A \cap B)$	P	$\left[\frac{2}{3}, 1\right]$
II	Range of $P(A \cup B)$	Q	$\left[\frac{4}{9}, 1\right]$
III	Range of $P(A / B)$	R	$\left[\frac{2}{5}, \frac{9}{10}\right]$
IV	Range of $P(B / A)$	S	$\left[\frac{4}{15}, \frac{3}{5}\right]$
		T	$\left[\frac{9}{10}, 1\right]$

	I	II	III	IV
(A)	S	P	R	Q
(B)	P	Q	R	S
(C)	S	P	Q	R
(D)	R	S	T	Q

53. Let two sets A and B be such that $n(A) = 10$, $n(B) = 12$ then

List -I		List -II	
I	$n(A \cup B)$ can be	P	2
II	$n(A \cap B)$ can be	Q	3
III	$n(A \Delta B)$ can be	R	13
IV	If $n(A \cap B) = 0$ then $n(A \cup B)$ is	S	16
		T	22

	I	II	III	IV
(A)	R, S, T	P, Q	P,Q,R,S,T	T
(B)	R, S, T	Q	P, Q	P
(C)	P, Q, S	P, Q	R, S, T	T

(D) R, S, T P, Q P, S, T T

54. $P(0, 3, -2)$, $Q(3, 7, -1)$ and $R(1, -3, -1)$ are 3 given points. Let L_1 be the line passing through P and Q and L_2 be the line through R and parallel to the vector $\hat{i} + \hat{k}$.

List -I		List -II	
I	Perpendicular distance of P from L_2	P	$7\sqrt{3}$
II	Shortest distance between L_1 and L_2	Q	2
III	Area of the ΔPQR	R	6
IV	Distance of P from centroid of triangle QRS, where $S(-4, -4, 2)$, is	S	$\sqrt{13}$
		T	$\sqrt{38}$

	I	II	III	IV
(A)	T	Q	S	P
(B)	R	P	Q	S
(C)	R	S	Q	P
(D)	R	Q	P	S

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